ClipSAT : Linear Algebra Vectors, Matrices, Eigenvalues & Least Squares

Unit 7.01: Linear Transformations & Change of Basis

Name: _____

Q1. Let Tmathbf x=Amathbf x with A=left[begin{matrix}-2 & -2-1 & 0end{matrix}right] . Find Tlangle 3, -3 rangle .

[ ] A: langle 0, -3 rangle
[ ] B: langle -3, -6 rangle
[ ] C: langle 0, 3 rangle
[ ] D: langle 1, -2 rangle

Q2. Which matrix represents the reflection across the line y = x?

[ ] A: left[begin{matrix}0 & -1-1 & 0end{ma
[ ] B: left[begin{matrix}0 & 11 & 0end{matr
[ ] C: left[begin{matrix}0 & -11 & 0end{mat
[ ] D: left[begin{matrix}1 & 00 & 1end{matr

Q3. Which map is linear?

[ ] A: Tx,y=x+1, y
[ ] B: Tx,y=xy, x
[ ] C: Tx,y=sin x, y
[ ] D: TA=A^T text{on } M_{2times2}

Q4. Let Tmathbf x=Amathbf x with A=left[begin{matrix}-3 & -12 & -2end{matrix}right] . Find Tlangle -3, 2 rangle .

[ ] A: langle 13, -1 rangle
[ ] B: langle -7, 10 rangle
[ ] C: langle 8, -9 rangle
[ ] D: langle 7, -10 rangle

Q5. A 3 times 3 symmetric matrix T_{a} has eigenvalues lambda_1 = 20 , lambda_2 = 22 , and lambda_3 = 20 . Find the exact value of the trace text{tr}T_{a} .

[ ] A: 62
[ ] B: 40
[ ] C: 82
[ ] D: 65

Q6. A linear map f: mathbb{R}^{20} to mathbb{R}^{20} has rank text{rank}f = 18 . Determine the exact value of the nullity dimker f .

[ ] A: 20
[ ] B: 5
[ ] C: 0
[ ] D: 2

Q7. Which matrix represents the orthogonal projection onto the x-axis?

[ ] A: left[begin{matrix}1 & 00 & 0end{matr
[ ] B: left[begin{matrix}-1 & 00 & 0end{mat
[ ] C: left[begin{matrix}0 & 00 & 1end{matr
[ ] D: left[begin{matrix}0 & -11 & 0end{mat

Q8. Which map is NOT linear?

[ ] A: Tx,y,z=x+z, 0
[ ] B: TA=det A text{on } M_{2times2}
[ ] C: Tp=p1 text{on } P_3
[ ] D: TA=A^T text{on } M_{2times2}